



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)
BSc CHEMISTRY**

Programme	B.Sc Chemistry				
Course Title	INORGANIC CHEMISTRY-II				
Type of Course	MAJOR				
Semester	4				
Academic Level	200-299				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	Classification of elements to different blocks, Comparative study of s and p block elements based on electronic configuration and atomic size, General idea about transition and inner transition elements, Concept of coordinate bond, Differences between double salts and complexes, Ligands, Coordination number. Concept of catenation and polymerization. Theoretical and practical knowledge about volumetric analysis.				
Course Summary	This course explains characteristics of s, p, d and f block elements. It also gives an insight into various theories in coordination compounds. It explores the application of inorganic chemistry in daily life. It covers practical application of complex formation in quantitative analysis.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	To Elucidate the trends in physical and chemical properties of s and p block elements	U	C	Instructor- created exams/ Quizzes/assignments
CO2	To Evaluate the general characteristics of Transition and Inner Transition elements, their comparison and applications	U	C	Instructor- created exams/ Quizzes/assignments

CO3	To demonstrate knowledge of coordination chemistry, isomerism and theories of bonding in coordination compounds	U	M	Instructor- created exams/ Quizzes/assignments
CO4	To analyze different types of inorganic polymers their structures, properties and applications	An	C	Instructor- created exams/ Quizzes/assignments
CO5	To Appreciate the utility of inorganic compounds in day to day life	Ap	M	Instructor- created exams/ Quizzes/assignments
CO6	To Apply the knowledge of complex formation and gain hands on experience in quantitative analysis with some day to day application	Ap	P	Group work /Viva Voce// Observation of practical skill

Detailed Syllabus:

Module	Unit	Content	Hrs	Mark
I	s & p BLOCK ELEMENTS		15	33
	1	s block General properties: Ionization Energy, Flame coloration, Photoelectric effect, Metallic character, Hydration energy.	2	
	2	p block elements: Comparative study- Halides, Sulfates, Carbonates and bicarbonates (solubility and thermal stability)	1	
	3	Oxidation number and inert pair effect, Comparison of Lewis acidity of boron halides.	2	
	4	Preparation, Properties, Structure and uses of Diborane, Boric acid, Borazine and Boron nitride, Structure of AlCl_3	3	
	5	Structure and bonding of oxides of N (N_2O , NO , N_2O_3 , NO_2 , N_2O_4 , N_2O_5) and S (SO_2 and SO_3)	2	
	6	Oxo acids of P (H_3PO_2 , H_3PO_3 , H_3PO_4) and Cl (HOCl , HOCl_2 , HOCl_3 , HOCl_4) (Structure and Acid strength), Colour and Bond Dissociation energy of halogens.	1	
	7	Interhalogen compounds: Preparation, Properties, Uses and Structure (One example each for AB , AB_3 , AB_5 and AB_7 types), Electropositive character of iodine, Pseudo	3	

		halogen: Comparison of Pseudo halogen (Cyanogen as example) and halogens and structure of Poly halide ions.		
	8	Noble gases: Isolation of noble gases: Dewar's method- Separation by charcoal adsorption method, Uses of He, and Ne	1	
II		TRANSITION AND INNER TRANSITION ELEMENTS	8hr	17
	9	Electronic configuration and General characteristics, Ionization energy, Colour, Magnetic properties, Reducing properties, Catalytic properties.	2	
	10	Non-stoichiometric compounds, Complex formation and Alloy formation. Comparison of 3d, 4d and 5d transition series. Important application of transition metals. Isopoly and heteropoly anions of W and Mo.	2	
	11	Lanthanides and Actinides- Electronic configuration and General properties. Isolation of Lanthanides from monazite sand, Separation by ion exchange method.	2	
	12	Magnetic properties. Lanthanide contraction, causes and consequences. Industrial importance of Lanthanides. Comparison of Actinides & Lanthanides [Mention only].	2	
III		COORDINATION CHEMISTRY	15 hr	33
	13	IUPAC Nomenclature of complexes, Types of ligands: (mono, bi, tri, tetra, hexa, ambidentate, chelate and macrocyclic ligands), Isomerism-Structural and Stereoisomerism,	2	
	14	Review of Werner's theory and Sidwick concept of coordination-EAN rule,	1	
	15	Factors affecting stability of complexes, Application of coordination complexes in quantitative and qualitative analysis.	2	
	16	Theories of bonding, VBT (valence bond theory), Geometry of coordination numbers 4 & 6, Limitation of VBT.	2	
	17	Crystal field Theory: CFT-splitting of d orbitals in Octahedral and Tetrahedral complexes. CFSE of low spin and high spin octahedral complexes- Normal and inverse	3	

		spinel compounds, Factors affecting crystal field splitting, Spectrochemical series.		
	18	CFT-splitting of d orbitals in Tetragonal and Square planar Complexes. Magnetism (spin only magnetic moment) and Colour (d-d transition), Distorted octahedral complexes-Jahn-Teller theorem, CFSE calculation and its applications, Merits and demerits of CFT.	5	
IV		INDUSTRIALLY IMPORTANT INORGANIC COMPOUNDS AND THEIR APPLICATION IN DAILY LIFE	7 hr	15
	19	Inorganic Polymers: Homochain Polymers and Heterochain Polymers.	1	
	20	Structure and Applications of Silicones, Silicates, Zeolites, Phosphazenes, Preparation, Properties and Structure of di and tri phosphonitrilic chlorides, SN compounds: Preparation Methods, Properties and Structure of S ₂ N ₂ , S ₄ N ₄ and (SN) _x ,	3	
	21	Refractory materials: Borides and Carbides, Inorganic fertilizers: Essential Nutrients to plants- Nitrogenous, Phosphate and Potash fertilizers-Examples with formula, Rocket Propellants: Classification with examples.	2	
	22	Cement: Ingredients, Setting of cement, Role of gypsum Glass: Varieties of glass.	1	
V		INORGANIC CHEMISTRY PRACTICAL II: COMPLEXOMETRIC TITRATIONS AND INORGANIC PREPARATIONS	30 hr	
		From Section A Minimum of 3 experiments must be done and from Section B Minimum 3 experiments must be done Section A Complexometry 1. Estimation of magnesium 2. Estimation of Zinc 3. Determination of hardness of water 4. Determination of COD of water samples Section B Inorganic preparations: (a) Ferric alum,		

		(b)Nickel (II) dimethylglyoximate, (c)Tetraammine copper (II) sulfate, (d)Potash alum Open Ended: Any two experiments related to complexometry can be selected by the teacher		
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References :

1. B.R. Puri L.R. Sharma, K.C. Kalia, Principles of Inorganic Chemistry, Milestone Publishers, New Delhi,2010.
2. S. Prakash, G. D. Tuli, S. K. Basu, R. D.Madan, Advanced Inorganic Chemistry,5th Edn., Vol.I,S Chand,2012.
3. J.D. Lee, Concise Inorganic Chemistry,5th Edn.,Wiley India Pvt.Ltd.,2008.
4. R. Gopalan, V.Ramalingam, Concise Coordination Chemistry, 1st Edn.,Vikas Publishing House, New Delhi,2001.
5. G. S. Manku ,Theoretical Principles of Inorganic Chemistry. McGraw-Hill Education; New edition (1 August 1982)
6. M.C. Day, J.Selbin,Theoretical Inorganic Chemistry,East West Press,New Delhi,2002.
7. J. E. Huheey, E.A.Keitler,R.L.Keitler,Inorganic Chemistry-Principles of Structure and Reactivity,4TH Edn.,Pearson Education, New Delhi,2013.
8. M.N. Greenwood, A. Earnshaw, Chemistry of elements, 2nd Edn., Butterworth,1997.
9. B.K. Sharma, Industrial chemistry, 11th Edn., Goel publishing House, Meerut, 2000.
10. J. Mendham, R. C. Denney, J. D. Barnes, M. Thomas, *Vogel's Text Book of Quantitative Chemical Analysis*, 6th Edn., Pearson Education, Noida, 2013.
11. D. A. Skoog, D. M. West, F. J. Holler, S. R. Crouch, *Fundamentals of Analytical Chemistry*, 8th Edn., Brooks/Cole, Thomson Learning, Inc., USA, 2004.

Further Reading

1. W.U.Malik, G.D.Tuli, R.D. Madan, selected Topics in Inorganic Chemistry, S. Chand and Co., New Delhi,2010(Reprint)
2. F.A.Cotton,G.Wilkinson,Advanced Inorganic Chemistry,6TH Edn.,Wiley India Pvt.Ltd., New Delhi,2009.
3. James E. House, Inorganic Chemistry, academic press, 2008.

Mapping of COs with PSOs and POs:

	PSO 1	PSO 2	PSO 3	PSO 4	PSO 5	PSO 6	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
C O 1	3			-	2	2	3				2		1
C O 2	3		-	-	2	2	3				1	1	2
C O 3	3	-		-	-	-	2		1		1		2
C O 4	3	-			3	3	3		2	1	2		2
C O 5	2		-	-	3	2	2		1	1	2	2	2
C O 6	2	-	2		3	3	2		2	1	3	1	2

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Discussion / Seminar
- Midterm Exam
- Practical exam (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment/ viva/seminar	Practical skill evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6		✓	✓	